Lecture Notes

Lecture 10

The Spectral Theorem

Lecture 9 showed that a normal operator has orthogonal eigenvectors for distinct eigenvalues. The *spectral theorem* completes the story in the most useful way possible: a normal operator on a complex space has a *complete* orthonormal basis of eigenvectors—it is diagonal in an orthonormal basis, i.e. *unitarily diagonalizable.* For observables (self-adjoint operators) the eigenvalues are moreover real. This single theorem is the mathematical backbone of quantum mechanics: every observable comes with a complete orthonormal set of eigenstates labelled by real measurement outcomes, and the measurement postulate, the Born rule, and unitary time evolution all rest on it. We then *perturb* that picture: if a solvable operator is nudged by a small self-adjoint εV, its spectrum cannot move far, and the shift can be computed order by order—the calculation every quantum mechanics course runs on.

Learning objectives

By the end of this lecture you will be able to:

- State the complex and real spectral theorems and the equivalences they assert.
- Diagonalize a Hermitian or normal operator by a unitary.
- Write the spectral decomposition $T = \sum_k \lambda_k P_k$ and compute $f(T)$ spectrally.
- Bound how far a perturbation moves a spectrum, and compute first- and second-order energy shifts.
- Relate commuting observables and simultaneous diagonalization to the measurement postulate.

1 Schur's theorem

The proof rests on triangularizing in an *orthonormal* basis.

Theorem 10.1 (Schur). Every operator on a finite-dimensional complex inner product space has an upper-triangular matrix with respect to some orthonormal basis.

Proof. By Lecture 5 (where triangularizability over $\mathbb{C}$ is sketched), there is *a* basis $v_1, \dots, v_n$ in which T is upper-triangular, i.e. each subspace $U_k = \operatorname{span}(v_1, \dots, v_k)$ is invariant, $TU_k \subseteq U_k$. Gram–Schmidt (Lecture 8) produces an orthonormal $e_1, \dots, e_n$ with $\operatorname{span}(e_1, \dots, e_k) = U_k$ for every k, so the same subspaces stay invariant; in this orthonormal basis T is upper-triangular. ∎

Example 10.2 (Triangularizing in an orthonormal basis). $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ has the single eigenvalue 1 with eigenvector $(0, 1)$; taking the orthonormal basis $e_1 = (0, 1)$, $e_2 = (1, 0)$ gives $\mathcal{M}(A) = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, upper-triangular as Schur promises—but not diagonal, since A is not normal. ◁

2 The complex spectral theorem

Theorem 10.3 (Complex spectral theorem). Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. The following are equivalent:

1. T is normal;
2. T has a diagonal matrix with respect to some orthonormal basis;
3. V has an orthonormal basis consisting of eigenvectors of T.

Proof. (2)$\Leftrightarrow$(3) is immediate, as the columns of a diagonal matrix in a basis say exactly that the basis vectors are eigenvectors. (2)$\Rightarrow$(1): a diagonal matrix and its conjugate transpose commute, so $T^\dagger T = TT^\dagger$.

(1)$\Rightarrow$(2): by Schur, take an orthonormal basis $e_1, \dots, e_n$ with $\mathcal{M}(T) = A$ upper-triangular. From the first column, $\|Te_1\|^2 = |a_{11}|^2$, while $\mathcal{M}(T^\dagger) = A^\dagger$ gives, from its first column (the conjugated first *row* of A), $\|T^\dagger e_1\|^2 = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2$. Normality forces $\|Te_1\| = \|T^\dagger e_1\|$ (Lecture 9), so $a_{12} = \cdots = a_{1n} = 0$. Repeating for each row in turn shows every off-diagonal entry vanishes, so A is diagonal. ■

In matrix language: A is normal iff $A = UDU^\dagger$ with U unitary and D diagonal. The columns of U are the orthonormal eigenvectors; conjugating by the unitary U (a change of orthonormal basis, Lecture 9) reveals the diagonal D.

$$\begin{tikzcd} V \arrow[r, "T"] & V \\ V \arrow[u, "U"] \arrow[r, "D = \mathrm{diag}(\lambda_k)"'] & V \arrow[u, "U"'] \end{tikzcd}$$

Figure 1: The spectral theorem: a normal operator is the diagonal D seen through a *unitary* change of orthonormal basis U, so $T = UDU^\dagger$. Compare the non-orthogonal $A = PDP^{-1}$ of Lecture 6.

Example 10.4 (Unitary diagonalization of a normal operator). The normal operator $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ of Lecture 9 has eigenvalues $1 \pm i$ with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, i)$ and $\frac{1}{\sqrt{2}}(1, -i)$. With these as the columns of a unitary U,

$$U^\dagger A U = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}.$$

The operator is diagonal in an orthonormal basis, exactly as the theorem guarantees—no eigenvalue need be real, only normal. ◁

Example 10.5 (The unitary that diagonalizes). For that same A, the orthonormal eigenvectors are the columns of $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$, and one verifies $U^\dagger A U = \mathrm{diag}(1+i, 1-i)$. The columns of U *are* the eigenstates, and $U^\dagger$ rotates the standard basis into the eigenbasis. ◁

Example 10.6 (Diagonalizing a unitary). The rotation $R = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ is normal, hence unitarily diagonalizable: in the orthonormal eigenbasis $\frac{1}{\sqrt{2}}(1, \mp i)$ it becomes

$$\mathrm{diag}(e^{i\theta}, e^{-i\theta}),$$

a diagonal of unit-modulus phases. A real rotation and a pair of conjugate phases are the same operator in two bases. ◁

Example 10.7 (Circulant matrices and the Fourier basis). A circulant matrix—cyclically constant along diagonals, e.g. $\left(\begin{smallmatrix} a & b & c \\ c & a & b \\ b & c & a \end{smallmatrix}\right)$—is normal, and *every* circulant is diagonalized by the one orthonormal Fourier basis $f_k = \frac{1}{\sqrt{n}}(1, \omega^k, \omega^{2k}, \dots)$ with $\omega = e^{2\pi i/n}$. Its eigenvalues are the discrete Fourier transform of the first row. This is why the FFT diagonalizes convolution: a single unitary serves all circulants at once. ◁

Remark 10.8 (The spectrum is a complete invariant). Two normal operators are unitarily equivalent if and only if they have the same eigenvalues with the same multiplicities, since each is unitarily equivalent to the diagonal matrix of its eigenvalues. For normal operators the spectrum (with multiplicity) is therefore a complete invariant—unlike the non-normal case, where Jordan structure also matters.

Remark 10.9 (Commuting normal operators). A commuting family of normal operators can be simultaneously unitarily diagonalized—a single orthonormal basis of joint eigenvectors serves all of them. This is the rigorous basis for the simultaneous measurement of compatible observables.

Remark 10.10 (Why normality is needed). Normality is not optional. The shear $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ is not normal and has only a one-dimensional eigenspace, so no basis—let alone an orthonormal one—consists of its eigenvectors. The spectral theorem is precisely the statement that normal is the exact dividing line for unitary diagonalizability.

3 Self-adjoint operators and the real case

> **Theorem 10.11 (Spectral theorem for self-adjoint operators).** A self-adjoint operator on a finite-dimensional inner product space (real or complex) has an orthonormal basis of eigenvectors, with all eigenvalues real. Equivalently, a Hermitian matrix factors as $A = UDU^\dagger$ with U unitary and D real diagonal; a real symmetric matrix factors as $A = ODO^\mathsf{T}$ with O orthogonal.

Proof. A self-adjoint operator is normal, so over $\mathbb{C}$ the complex spectral theorem gives an orthonormal eigenbasis, and the eigenvalues are real by Lecture 9.

Over $\mathbb{R}$ the argument needs one extra step, because Theorem 10.1 as stated covers complex spaces. Let A be the real symmetric matrix of T in an orthonormal basis and regard it for a moment as a complex Hermitian matrix. Its characteristic polynomial is the same either way, and by Lecture 9 all of its roots are real; hence χ_T splits over $\mathbb{R}$. The proof of Theorem 10.1 uses nothing about $\mathbb{C}$ beyond the fact that χ_T splits—it triangularizes and then runs Gram–Schmidt—so it applies verbatim over $\mathbb{R}$ and yields a real orthonormal basis in which T is upper-triangular. A real symmetric upper-triangular matrix equals its transpose, hence is diagonal: an orthogonal eigenbasis with real entries. ∎

Physically this is the crucial statement: *every observable has a complete orthonormal set of eigenstates with real eigenvalues.* The real version is the "principal axes theorem"—a real symmetric matrix (inertia tensor, stress tensor, quadratic form) is diagonal in a suitable set of mutually perpendicular axes.

Example 10.12 (A Hermitian 2×2). $A = \left(\begin{smallmatrix}2&i\\-i&2\end{smallmatrix}\right)$ is Hermitian, with real eigenvalues $1, 3$ and orthonormal eigenvectors $e_1 = \frac{1}{\sqrt{2}}(1, i)$, $e_2 = \frac{1}{\sqrt{2}}(1, -i)$. With $U = (e_1\, e_2)$, $U^\dagger AU = \operatorname{diag}(1, 3)$: a two-state observable in its eigenbasis. ◁

Example 10.13 (Principal axes of a quadratic form). The quadratic form $q(x, y) = 2x^2 + 2y^2 + 2xy$ has matrix $\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$, with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, 1)$, $\frac{1}{\sqrt{2}}(1, -1)$ and eigenvalues $3, 1$. In the rotated coordinates (x', y') along those axes, $q = 3x'^2 + y'^2$: a real symmetric matrix is diagonalized by a rotation, straightening a tilted ellipse into one aligned with the axes. ◁

Example 10.14 (A degenerate self-adjoint operator). $A = \mathrm{diag}(4, 4, 1)$ on $\mathbb{R}^3$ is self-adjoint with eigenvalue 4 (multiplicity 2) and 1. The 4-eigenspace is the whole xy-plane, so *any* orthonormal basis of that plane, together with e_3, diagonalizes A. Degeneracy leaves freedom in the eigenbasis—freedom fixed in physics by a further commuting observable. ◃

Example 10.15 (Normal modes from the real spectral theorem). The stiffness matrix $K = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ of two coupled masses is real symmetric, so the real spectral theorem furnishes an *orthogonal* eigenbasis: $\frac{1}{\sqrt{2}}(1, 1)$ with $\omega^2 = 1$ and $\frac{1}{\sqrt{2}}(1, -1)$ with $\omega^2 = 3$. In these normal coordinates the coupled system $\ddot{x} = -Kx$ decouples into independent oscillators (Lecture 6)—the modes are perpendicular precisely because K is self-adjoint. ◃

Example 10.16 (Positive operators have nonnegative spectrum). $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ is self-adjoint with eigenvalues $3, 1 > 0$, hence *positive*: in eigenbasis coordinates $\langle v, Av\rangle = 3|c_1|^2 + |c_2|^2 \geq 0$. Positive operators are exactly the self-adjoint ones with nonnegative spectrum—the operator analogue of the nonnegative reals, and the ones with real square roots. ◃

4 Spectral decomposition

Group the eigenvalues into the distinct values $\lambda_1, \dots, \lambda_m$, and let P_k be the orthogonal projection onto the eigenspace $E(\lambda_k, T)$.

> **Theorem 10.17 (Spectral decomposition).** A normal operator T (on a complex space; or self-adjoint on a real space) with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ satisfies
>
> $$P_j P_k = \delta_{jk} P_k, \qquad P_1 + \cdots + P_m = I, \qquad T = \lambda_1 P_1 + \cdots + \lambda_m P_m,$$
>
> with each P_k self-adjoint ($P_k^\dagger = P_k$). Consequently $f(T) = \sum_k f(\lambda_k) P_k$ for any function f on the spectrum.

Proof. The eigenspaces are mutually orthogonal (Lecture 9) and, by the spectral theorem, span V, so $V = E(\lambda_1) \oplus \cdots \oplus E(\lambda_m)$ orthogonally. The P_k are then orthogonal projections onto orthogonal summands, giving $P_j P_k = \delta_{jk} P_k$ and $\sum_k P_k = I$. Applying both sides of the last equation to an eigenvector shows $T = \sum_k \lambda_k P_k$, and orthogonal projections are self-adjoint. Applying f eigenvalue by eigenvalue gives $f(T) = \sum_k f(\lambda_k) P_k$. ■

In Dirac notation, with an orthonormal eigenbasis $\{|e_k\rangle\}$,

$$T = \sum_k \lambda_k |e_k\rangle\langle e_k|, \qquad I = \sum_k |e_k\rangle\langle e_k| :$$

the operator is a weighted sum of the rank-one projectors $|e_k\rangle\langle e_k|$, and the identity is their sum (the completeness relation of Lecture 8). This is the form physicists mean by "diagonalizing an observable."

Example 10.18 (Spectral decomposition of σ_z). $\sigma_z = \mathrm{diag}(1, -1)$ has eigenprojections $P_+ = |0\rangle\langle 0| = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $P_- = |1\rangle\langle 1| = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, with $P_+ + P_- = I$, $P_+ P_- = 0$, and $\sigma_z = (+1)P_+ + (-1)P_-$. Any function follows at once: $e^{i\theta\sigma_z} = e^{i\theta} P_+ + e^{-i\theta} P_- = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. ◃

Example 10.19 (Spectral decomposition of σ_x). σ_x has eigenvalues ± 1 with projectors $P_\pm = |\pm\rangle\langle\pm| = \frac{1}{2}\left(\begin{smallmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{smallmatrix}\right)$, so $\sigma_x = P_+ - P_-$ and $f(\sigma_x) = f(1)P_+ + f(-1)P_-$ for any f. Taking $f(\lambda) = e^{i\theta\lambda}$ recovers $e^{i\theta\sigma_x} = \cos\theta\, I + i \sin\theta\, \sigma_x$ from Lecture 6, now read straight off the spectral form. ◃

Example 10.20 (A function via the spectral form). For the Hermitian A of Example 10.12, with eigenprojections P_1, P_3 onto the eigenvalue-1 and eigenvalue-3 spaces, $\sqrt{A} = P_1 + \sqrt{3}\, P_3$ and

$e^A = e\,P_1 + e^3 P_3$. Because A is positive, $\sqrt{A}$ is a genuine positive square root—the spectral calculus of Lecture 6 in orthonormal form. ◃

Example 10.21 (A positive square root, explicitly). The positive operator $A = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 5 \end{smallmatrix}\right)$ has eigenvalues $8, 2$ with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, \pm 1)$, so $P_\pm = \frac{1}{2}(I \pm \sigma_x)$ and

$$\sqrt{A} = \sqrt{8}\,P_+ + \sqrt{2}\,P_- = \frac{\sqrt{8}+\sqrt{2}}{2}\,I + \frac{\sqrt{8}-\sqrt{2}}{2}\,\sigma_x,$$

the unique positive operator squaring to A. ◃

Example 10.22 (Eigenprojections written out). For $A = \left(\begin{smallmatrix} 2 & i \\ -i & 2 \end{smallmatrix}\right)$ of Example 10.12, the eigenprojections are

$$P_1 = |e_1\rangle\langle e_1| = \tfrac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}, \qquad P_3 = |e_2\rangle\langle e_2| = \tfrac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix},$$

which satisfy $P_1 + P_3 = I$, $P_1 P_3 = 0$, and $A = 1 \cdot P_1 + 3 \cdot P_3$. ◃

Remark 10.23 (Projections by interpolation). The eigenprojections can be written without finding eigenvectors: $P_k = \prod_{j \neq k} \dfrac{A - \lambda_j I}{\lambda_k - \lambda_j}$, the Lagrange polynomial equal to 1 at λ_k and 0 at the other eigenvalues. For σ_z this gives $P_\pm = \frac{1}{2}(I \pm \sigma_z)$ at once.

Example 10.24 (Spectral decomposition of σ_y). σ_y has eigenvalues ± 1 with eigenvectors $\frac{1}{\sqrt{2}}(1, \pm i)$, hence projectors $P_\pm = \frac{1}{2}\left(\begin{smallmatrix} 1 & \mp i \\ \pm i & 1 \end{smallmatrix}\right)$ and $\sigma_y = P_+ - P_-$. The three Pauli operators share the eigenvalues ± 1 but have mutually different eigenprojections—the hallmark of incompatible observables. ◃

Example 10.25 (Reconstructing an operator). Conversely, spectral data determine the operator: from eigenvalues $2, 5$ with projectors $P_2 = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $P_5 = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$,

$$A = 2P_2 + 5P_5 = \tfrac{1}{2}\begin{pmatrix} 7 & -3 \\ -3 & 7 \end{pmatrix}.$$

The operator and its spectral data are interchangeable. ◃

Example 10.26 (A projector resolution in $\mathbb{C}^3$). For $A = \operatorname{diag}(2, 2, 5)$ on $\mathbb{C}^3$ the eigenprojections are $P_2 = \operatorname{diag}(1, 1, 0)$ and $P_5 = \operatorname{diag}(0, 0, 1)$, with $P_2 + P_5 = I$, $P_2 P_5 = 0$, and $A = 2P_2 + 5P_5$. The degenerate eigenvalue 2 carries a rank-2 projector, and $\operatorname{tr} P_2 = 2$ counts its multiplicity. ◃

Example 10.27 (Inverse from the spectrum). A normal $T = \sum_k \lambda_k P_k$ with every $\lambda_k \neq 0$ is invertible, with $T^{-1} = \sum_k \lambda_k^{-1} P_k$: indeed $\left(\sum_k \lambda_k P_k\right)\left(\sum_j \lambda_j^{-1} P_j\right) = \sum_k P_k = I$, using $P_j P_k = \delta_{jk} P_k$. No function of an operator asks more than the same function of its eigenvalues. ◃

Remark 10.28 (The functional calculus). The map $f \mapsto f(T)$ is an algebra homomorphism: $(f+g)(T) = f(T) + g(T)$, $(fg)(T) = f(T)g(T)$, and $\overline{f}(T) = f(T)^\dagger$. Because $f(T) = \sum_k f(\lambda_k) P_k$, each of these reduces to the corresponding fact about numbers applied eigenvalue by eigenvalue—the *functional calculus* of a normal operator.

One more quantity is read straight off the spectrum. How much can an operator stretch a vector?

> **Definition 10.29 (Operator norm).** For $T \in \mathcal{L}(V)$ on an inner product space, the *operator norm* is $\|T\| := \sup_{v \neq 0} \|Tv\|/\|v\| = \max_{\|v\|=1} \|Tv\|$, finite in finite dimensions (a continuous function maximized on the unit sphere). At once $\|Tv\| \leq \|T\|\,\|v\|$, and hence

$\|ST\| \le \|S\| \, \|T\|$.

Proposition 10.30 (Norm and resolvent of a normal operator). Let $T = \sum_k \lambda_k P_k$ be normal. Then

$$\|T\| = \max_k |\lambda_k|,$$

and for every $z \notin \operatorname{spec}(T)$ the operator $T - zI$ is invertible with

$$\|(T - zI)^{-1}\| = \frac{1}{\operatorname{dist}\big(z, \operatorname{spec}(T)\big)}, \qquad \operatorname{dist}\big(z, \operatorname{spec}(T)\big) = \min_k |\lambda_k - z|.$$

Proof. Resolve $v = \sum_k P_k v$ into orthogonal eigen-components. By Pythagoras (Lecture 7), $\|v\|^2 = \sum_k \|P_k v\|^2$ and

$$\|Tv\|^2 = \sum_k |\lambda_k|^2 \|P_k v\|^2 \le \Big(\max_k |\lambda_k|\Big)^2 \|v\|^2,$$

with equality when v is an eigenvector for a largest-modulus eigenvalue; hence $\|T\| = \max_k |\lambda_k|$. For the second claim, the functional calculus applied to $f(\lambda) = (\lambda - z)^{-1}$ gives the normal operator $(T - zI)^{-1} = \sum_k (\lambda_k - z)^{-1} P_k$, whose norm is therefore $\max_k |\lambda_k - z|^{-1} = \big(\min_k |\lambda_k - z|\big)^{-1}$. ∎

The resolvent $(T - zI)^{-1}$ thus blows up exactly as z approaches the spectrum, at a rate set by nothing but the distance to the nearest eigenvalue.

Remark 10.31 (Trace and determinant from the spectrum). In the eigenbasis $\operatorname{tr} T = \sum_k \lambda_k$ and $\det T = \prod_k \lambda_k$, both basis-independent (Lecture 5). The trace of an eigenprojection P_k is the dimension of its eigenspace, so $\operatorname{tr} T = \sum_k \lambda_k \operatorname{tr} P_k$ recovers the trace directly from the spectral decomposition.

Remark 10.32 (Singular value decomposition). Even a non-normal T yields to a spectral idea: applying the theorem to the positive operator $T^\dagger T$ gives the *singular value decomposition* $T = U\Sigma V^\dagger$ with U, V unitary and Σ diagonal with nonnegative entries—the universal factorization behind low-rank approximation and the pseudoinverse.

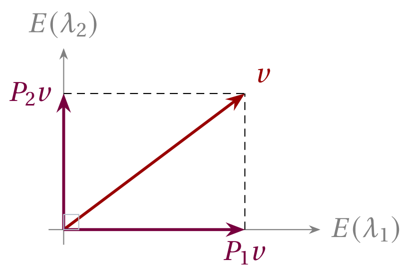

Figure 2: Spectral decomposition: the orthogonal eigenspaces split V into perpendicular summands, and $v = P_1 v + P_2 v + \cdots$ resolves every vector into its eigen-components. The observable acts on each piece by the scalar λ_k.

5 Perturbing a spectral decomposition

Exact diagonalization is the exception, not the rule. Almost every operator a physicist actually meets has the shape $H(\varepsilon) = H_0 + \varepsilon V$, with H_0 self-adjoint and already diagonalized, V self-adjoint, and $\varepsilon \in \mathbb{R}$ small: a spin in a slightly tilted field, an atom in a weak electric field, a lattice with one defective site. Two questions, in this order. How far can the spectrum move at all? And, knowing it moves only a little, what is the shift to leading orders in ε? Throughout, a denominator is written with the level of interest first, $E_n - E_m$.

How far can the spectrum move?

The first question has a clean answer, with Proposition 10.30 as the engine. It needs one standard tool, the operator version of $\frac{1}{1-x} = 1 + x + x^2 + \cdots$.

Lemma 10.33 (Neumann series). If $X \in \mathcal{L}(V)$ satisfies $\|X\| < 1$, then $I - X$ is invertible and

$$(I - X)^{-1} = \sum_{n \geq 0} X^n, \qquad \|(I - X)^{-1}\| \leq \frac{1}{1 - \|X\|}.$$

Proof. Sub-multiplicativity (Definition 10.29) gives $\|X^n\| \leq \|X\|^n$, so the series converges absolutely in the finite-dimensional space $\mathcal{L}(V)$. Its partial sums satisfy $(I - X)S_N = S_N(I - X) = I - X^{N+1} \to I$, since $\|X^{N+1}\| \leq \|X\|^{N+1} \to 0$; hence the limit S obeys $(I - X)S = S(I - X) = I$, with $\|S\| \leq \sum_n \|X\|^n = (1 - \|X\|)^{-1}$. ■

Theorem 10.34 (Spectral stability). Let H_0 and V be self-adjoint, $\varepsilon \in \mathbb{R}$, and let $E_1, \ldots, E_r$ be the distinct eigenvalues of H_0. Then

$$\operatorname{spec}\bigl(H_0 + \varepsilon V\bigr) \subseteq \bigcup_{k=1}^{r} \bigl\{z \in \mathbb{C} : |z - E_k| \leq |\varepsilon|\, \|V\|\bigr\}.$$

Proof. Let z satisfy $\operatorname{dist}(z, \operatorname{spec} H_0) > |\varepsilon|\|V\|$; we show $H_0 + \varepsilon V - zI$ is invertible. First, $z \notin \operatorname{spec}(H_0)$, so $H_0 - zI$ is invertible, and we may factor

$$H_0 + \varepsilon V - zI = (H_0 - zI)\bigl(I + \varepsilon(H_0 - zI)^{-1}V\bigr).$$

Since H_0 is normal, Proposition 10.30 and sub-multiplicativity give

$$\bigl\|-\varepsilon(H_0 - zI)^{-1}V\bigr\| \leq |\varepsilon|\, \|(H_0 - zI)^{-1}\|\, \|V\| = \frac{|\varepsilon|\, \|V\|}{\operatorname{dist}(z, \operatorname{spec} H_0)} < 1,$$

so the second factor is invertible by Lemma 10.33. A product of invertible operators is invertible, so $z \notin \operatorname{spec}(H_0 + \varepsilon V)$. ■

Every eigenvalue therefore stays within $|\varepsilon|\|V\|$ of an unperturbed one: the spectrum is *Lipschitz* in the perturbation—no smallness assumption, no series. The work was done by the resolvent bound of Proposition 10.30, which is exactly where self-adjointness entered.

Proposition 10.35 (No level is lost·cited). If $2|\varepsilon|\|V\| < \min_{j \neq k}|E_j - E_k|$, the disks of Theorem 10.34 are pairwise disjoint, and each contains exactly as many eigenvalues of $H_0 + \varepsilon V$, with multiplicity, as H_0 has at its centre.

A small perturbation thus moves levels without destroying them or letting two groups merge, provided the disks stay apart. The controlling ratio is *perturbation size over spectral gap*—a theme that returns in every formula below; Example 10.40 works it out.

First- and second-order corrections

Now assume every eigenvalue of H_0 is *simple* (non-degenerate), with $H_0|m\rangle = E_m|m\rangle$ for an orthonormal basis $\{|m\rangle\}$ and eigenprojections $P_m = |m\rangle\langle m|$. Fix a level $|n\rangle$.

> **Theorem 10.36 (Analyticity of a simple level · sketched).** If E_n is a simple eigenvalue of H_0, then for small $|\varepsilon|$ the operator $H(\varepsilon)$ has exactly one eigenvalue $E_n(\varepsilon)$ near E_n, and it—with a suitable choice of eigenvector—is given by a convergent power series in ε.

Sketch. Uniqueness is Proposition 10.35. For the series, apply the implicit function theorem to $p(\lambda, \varepsilon) = \det\big(\lambda I - H(\varepsilon)\big)$, a polynomial in both arguments: $p(E_n, 0) = 0$ and $\partial p/\partial\lambda \neq 0$ there, *precisely because* E_n is a simple root, giving a unique analytic branch through E_n. That settles the eigenvalue but says nothing about the eigenvector, so apply the same theorem a second time, to the pair

$$F(v, \lambda, \varepsilon) = \big(H(\varepsilon)v - \lambda v,\ \langle n \mid v\rangle - 1\big),$$

which vanishes at $(|n\rangle, E_n, 0)$. Its derivative there in (v, λ) sends (w, μ) to $\big((H_0 - E_n)w - \mu|n\rangle,\ \langle n \mid w\rangle\big)$; pairing the first component with $\langle n|$ kills the left-hand side and forces $\mu = 0$, after which simplicity of E_n gives $w \in \operatorname{span}(|n\rangle)$ and $\langle n \mid w\rangle = 0$ gives $w = 0$. The derivative is therefore injective, hence invertible, and $|n(\varepsilon)\rangle$ is analytic as well—normalized, as it happens, exactly as (10.1) requires. ■

This licenses the ansatz; without it what follows would be formal. Write

$$E_n(\varepsilon) = E_n + \varepsilon E_n^{(1)} + \varepsilon^2 E_n^{(2)} + \cdots, \qquad |n(\varepsilon)\rangle = |n\rangle + \varepsilon|n^{(1)}\rangle + \varepsilon^2|n^{(2)}\rangle + \cdots, \tag{10.1}$$

normalizing by $\langle n \mid n(\varepsilon)\rangle = 1$, i.e. $\langle n \mid n^{(k)}\rangle = 0$ for $k \geq 1$ (the eigenvector is fixed only up to scale; this is the convenient choice). Substituting into $H(\varepsilon)|n(\varepsilon)\rangle = E_n(\varepsilon)|n(\varepsilon)\rangle$, the terms of order ε and ε^2 read

$$(H_0 - E_n)|n^{(1)}\rangle = \big(E_n^{(1)} - V\big)|n\rangle, \qquad (H_0 - E_n)|n^{(2)}\rangle = \big(E_n^{(1)} - V\big)|n^{(1)}\rangle + E_n^{(2)}|n\rangle. \tag{10.2}$$

Pair the first with $\langle n|$: the left side vanishes, since H_0 is self-adjoint with E_n real, so $\langle n|\,(H_0 - E_n)\,|n^{(1)}\rangle = \langle (H_0 - E_n)n, n^{(1)}\rangle = 0$, while the right side is $E_n^{(1)} - \langle n|\,V\,|n\rangle$. Pairing it with $\langle m|$ for $m \neq n$ gives $(E_m - E_n)\langle m \mid n^{(1)}\rangle = -\langle m|\,V\,|n\rangle$. Pairing the second with $\langle n|$ kills its left side for the same reason, leaving $0 = -\langle n|\,V\,|n^{(1)}\rangle + E_n^{(2)}$. Together:

$$\boxed{E_n^{(1)} = \langle n|\,V\,|n\rangle, \qquad |n^{(1)}\rangle = \sum_{m\neq n} \frac{\langle m|\,V\,|n\rangle}{E_n - E_m}|m\rangle, \qquad E_n^{(2)} = \sum_{m\neq n} \frac{|\langle m|\,V\,|n\rangle|^2}{E_n - E_m}} \tag{10.3}$$

(the last using $\langle n|\,V\,|m\rangle = \overline{\langle m|\,V\,|n\rangle}$, since V is self-adjoint). These are the *Rayleigh–Schrödinger* formulas: a level shifts by the diagonal matrix element of V in its own state, the state picks up an admixture of every other state weighted by coupling over gap, and at second order the level shifts by a sum over all other levels, coupling squared over gap.

Remark 10.37 (The reduced resolvent). Formula (10.3) looks basis-dependent but is not. With the *reduced resolvent* $S_n := \sum_{m\neq n} P_m/(E_n - E_m)$—the inverse of $E_n - H_0$ on the orthogonal complement of $|n\rangle$—it reads

$$|n^{(1)}\rangle = S_n V|n\rangle, \qquad E_n^{(2)} = \langle n|\,V S_n V\,|n\rangle,$$

written entirely in the eigenprojections of Theorem 10.17. This is the form that survives into infinite dimensions (Lecture 14), where S_n is built from the resolvent $(H_0 - z)^{-1}$ rather than from a finite sum.

Remark 10.38 (Level repulsion, and two free checks). Every denominator in $E_n^{(2)}$ is negative for the *lowest* level and positive for the highest, so second order always pushes the ground state down and the top state up: levels repel. Two identities check any calculation—summing the first of (10.3) gives $\sum_n E_n^{(1)} = \operatorname{tr} V$, and pairing the (m,n) and (n,m) terms of the third, whose denominators differ only in sign, gives $\sum_n E_n^{(2)} = 0$. Both are forced by $\operatorname{tr} H(\varepsilon) = \operatorname{tr} H_0 + \varepsilon \operatorname{tr} V$ being exactly linear in ε.

Remark 10.39 (When is the perturbation small?). The expansion parameter is not ε but the dimensionless ratio $\varepsilon|\langle m| V |n\rangle|/|E_n - E_m|$: coupling measured against the *gap*. A nominally tiny ε buys nothing when two levels are nearly degenerate—the situation the next subsection addresses. (The companion identity $dE_n/d\varepsilon = \langle n(\varepsilon)| V |n(\varepsilon)\rangle/\langle n(\varepsilon) \mid n(\varepsilon)\rangle$, the *Hellmann–Feynman* theorem, is set as an exercise. The denominator is 1 for a unit eigenvector, but not for the intermediate-normalized $|n(\varepsilon)\rangle$ of (10.1), whose norm is $1 + O(\varepsilon^2)$.)

Example 10.40 (A three-level system, worked out). Take, in the standard basis $|1\rangle, |2\rangle, |3\rangle$,

$$H_0 = \operatorname{diag}(0,1,3), \qquad V = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -3 \end{pmatrix}.$$

First, stability. The eigenvalues of V are $-3.3460,\ 0.4067,\ 2.9392$, so $\|V\| = 3.3460$ by Proposition 10.30, while the smallest gap of H_0 is 1. Proposition 10.35 therefore applies while $2|\varepsilon|(3.3460) < 1$, i.e. $|\varepsilon| < 0.149$: up to there the three levels sit in three disjoint disks and can be tracked one by one.

Now the shifts. To first order they are the diagonal entries of V,

$$E_1^{(1)} = 2, \qquad E_2^{(1)} = 1, \qquad E_3^{(1)} = -3,$$

summing to $0 = \operatorname{tr} V$ as they must. All off-diagonal entries equal 1, so the second-order shifts are

$$E_1^{(2)} = \tfrac{1}{0-1} + \tfrac{1}{0-3} = -\tfrac{4}{3}, \qquad E_2^{(2)} = \tfrac{1}{1-0} + \tfrac{1}{1-3} = \tfrac{1}{2}, \qquad E_3^{(2)} = \tfrac{1}{3-0} + \tfrac{1}{3-1} = \tfrac{5}{6},$$

summing to $-\tfrac{4}{3} + \tfrac{1}{2} + \tfrac{5}{6} = 0$, the second check. Read physically, the lowest level is pushed *down* by both partners, the highest *up* by both, and the middle one, with a partner on each side, barely moves. For the ground state $E_1(\varepsilon) \approx 2\varepsilon - \tfrac{4}{3}\varepsilon^2$, against the exact eigenvalue:

ε	first order	second order	exact
0.05	0.1	0.096667	0.096562
0.10	0.2	0.186667	0.185892
0.20	0.4	0.346667	0.341699

Doubling ε multiplies the first-order error by about 4 and the second-order error by about 8—the errors are $O(\varepsilon^2)$ and $O(\varepsilon^3)$, as (10.1) predicts—and the accuracy visibly deteriorates at $\varepsilon = 0.2$, just past the 0.149 beyond which the disks overlap and the levels can no longer be tracked separately. The first-order *state* is equally concrete: (10.3) gives $|1(\varepsilon)\rangle \approx (1, -0.1, -0.0333)$ at $\varepsilon = 0.1$. Scaled to the same intermediate normalization $\langle 1 \mid 1(\varepsilon)\rangle = 1$, the exact eigenvector is $(1, -0.1055, -0.0356)$—the leading component agrees identically, by the normalization, and the two admixtures are right to within about 6%. (Its unit-normalized form is $(0.9939, -0.1049, -0.0354)$; comparing against *that* would charge the approximation for a change of normalization.) ◁

Degenerate levels *[self-study]*

Formula (10.3) divides by zero the moment two levels coincide, and Theorem 10.36 does not apply, having needed a *simple* root. This is no technical blemish: a degenerate level does not shift, it *splits. The remainder of this section is offered for self-study; Lectures 11–14 do not depend on it.*

The resolution is a reading of Example 10.14, where a degenerate eigenvalue was seen to leave freedom in the eigenbasis—freedom fixed in physics by a further commuting observable. A perturbation fixes it too, and in the only way it can.

> **Theorem 10.41 (First-order splitting · sketched).** Let E_0 be an eigenvalue of H_0 of multiplicity d, with eigenprojection P. To first order the level splits into $E_0+\varepsilon\mu_1,\dots,E_0+\varepsilon\mu_d$, where the μ_j are the eigenvalues of the *compressed* operator PVP restricted to range P—a $d\times d$ self-adjoint matrix—and the eigenvectors of PVP are the correct zeroth-order states.

Sketch. Granting that the perturbed eigenvalue and eigenvector admit expansions (10.1) about some $|\psi\rangle \in$ range P (for a self-adjoint family this is Rellich's theorem, quoted here), the order-ε equation reads $(H_0-E_0)|\psi^{(1)}\rangle = \big(E^{(1)}-V\big)|\psi\rangle$. Apply P: since H_0 commutes with P and $(H_0-E_0)P=0$, the left side vanishes, leaving $0 = E^{(1)}|\psi\rangle - PVP|\psi\rangle$ (using $P|\psi\rangle = |\psi\rangle$). So $|\psi\rangle$ is an eigenvector of PVP on range P with eigenvalue $E^{(1)}$; and PVP being self-adjoint on the d-dimensional range P, the spectral theorem supplies d orthonormal such states with real eigenvalues. ■

In one sentence: *a degeneracy is lifted by the part of the perturbation acting inside the degenerate eigenspace*, and the correct zeroth-order states are those diagonalizing it—which is why a physicist picks "good quantum numbers" to diagonalize the perturbation before perturbing. Zeeman and Stark splitting are precisely this calculation.

Example 10.42 (A degeneracy split). Take $H_0=\mathrm{diag}(1,1,4)$ and V with rows $(0,2,1)$, $(2,0,1)$, $(1,1,0)$. The non-degenerate formula is meaningless here: $\langle 1|V|2\rangle = 2 \neq 0$ while $E_1-E_2=0$. Instead compress V onto range $P=\mathrm{span}(|1\rangle,|2\rangle)$:

$$PVP\big|_{\mathrm{range}\,P} = \begin{pmatrix}0 & 2\\ 2 & 0\end{pmatrix} = 2\sigma_x, \qquad \mu=\pm 2,$$

so the degenerate level splits as $E \approx 1\pm 2\varepsilon$, with correct zeroth-order states $|\pm\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |2\rangle)$—the σ_x eigenstates of Example 10.19. At $\varepsilon = 0.1$ the exact eigenvalues are $0.800000,\ 1.192875,\ 4.007125$ against the predicted $0.8,\ 1.2,\ 4$.

Two things are worth noticing. The lower branch $1-2\varepsilon$ is *exact to all orders*, for a reason worth more than any series: $\langle 3|V|-\rangle = 0$, so $|-\rangle$ is an exact eigenvector of $H(\varepsilon)$ for every ε. And once the degeneracy is lifted the levels are simple again, so ordinary second-order theory resumes—for the upper branch only $|3\rangle$ is left to sum over, giving $E_+^{(2)} = |\langle 3|V|+\rangle|^2/(1-4) = -\frac{2}{3}$ and $1+2\varepsilon-\frac{2}{3}\varepsilon^2 = 1.193333$ at $\varepsilon=0.1$, against the exact 1.192875. ◁

6 Measurement and what comes next

The spectral theorem is the measurement postulate in mathematical dress. An observable $A=\sum_k \lambda_k P_k$ can return only its real eigenvalues λ_k; a state $|\psi\rangle$ yields outcome λ_k with probability $\|P_k|\psi\rangle\|^2$ (the Born rule), after which the state collapses to the normalized projection $P_k|\psi\rangle/\|P_k|\psi\rangle\|$. Because $\sum_k P_k = I$ and the P_k are orthogonal, these probabilities sum to one. Expectation values are $\langle A\rangle = \sum_k \lambda_k \|P_k|\psi\rangle\|^2$.

Example 10.43 (A measurement). Measuring σ_z on $|\psi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2|1\rangle)$: the outcome $+1$ occurs with probability $\|P_+|\psi\rangle\|^2 = \frac{1}{5}$ and -1 with probability $\frac{4}{5}$, collapsing $|\psi\rangle$ to $|0\rangle$ or $|1\rangle$ respectively. The expectation value is $\langle\sigma_z\rangle = (+1)\frac{1}{5} + (-1)\frac{4}{5} = -\frac{3}{5}$, the probability-weighted average of the eigenvalues. ◁

Example 10.44 (Measuring a degenerate observable). For $A = \operatorname{diag}(5,5,2)$ on $\mathbb{C}^3$ the eigenvalue 5 has a two-dimensional eigenspace with projection $P_5 = \operatorname{diag}(1,1,0)$. Measuring A on $|\psi\rangle = (a,b,c)$ returns 5 with probability $\|P_5\psi\|^2 = |a|^2 + |b|^2$, collapsing the state into the 5-eigenspace rather than to a single vector. Degenerate outcomes project onto whole eigenspaces. ◁

Example 10.45 (Time evolution via the spectral form). If $\hat{H} = \sum_n E_n P_n$ then

$$e^{-i\hat{H}t/\hbar} = \sum_n e^{-iE_n t/\hbar} P_n,$$

so evolution acts on each energy eigenspace by a phase. A state $|\psi\rangle = \sum_n P_n|\psi\rangle$ becomes $\sum_n e^{-iE_n t/\hbar} P_n|\psi\rangle$, and the outcome probabilities $\|P_n|\psi\rangle\|^2$ are conserved—energy conservation, read off the spectral decomposition. ◁

Example 10.46 (Measuring in a rotated basis). Prepare $|0\rangle$ (a σ_z-eigenstate) and measure σ_x, whose eigenstates are $|\pm\rangle$. The amplitudes $\langle +,0\rangle = \langle -,0\rangle = \frac{1}{\sqrt{2}}$ give both outcomes ± 1 probability $\frac{1}{2}$: a state sharp in one observable is maximally uncertain in an incompatible one. ◁

Remark 10.47 (Schrödinger and Heisenberg pictures). One may evolve states by $e^{-i\hat{H}t/\hbar}$ or, equivalently, evolve observables by $A(t) = e^{i\hat{H}t/\hbar} A e^{-i\hat{H}t/\hbar}$. Conjugation by a unitary preserves the spectrum, so $A(t)$ has the same eigenvalues as A at all times: the possible outcomes never change, only their probabilities.

Example 10.48 (A three-outcome measurement). Measuring $A = \operatorname{diag}(1,2,3)$ on $|\psi\rangle = \frac{1}{\sqrt{3}}(1,1,1)$ gives each outcome $1,2,3$ with probability $\frac{1}{3}$, so

$$\langle A\rangle = \tfrac{1}{3}(1+2+3) = 2, \qquad \langle A^2\rangle = \tfrac{1}{3}(1+4+9) = \tfrac{14}{3},$$

and the variance is $\langle A^2\rangle - \langle A\rangle^2 = \frac{14}{3} - 4 = \frac{2}{3}$. The spectral data deliver every statistical moment of a measurement. ◁

Remark 10.49 (Repeated measurement). Immediately repeating a measurement returns the same outcome with certainty: the first measurement leaves the state in an eigenspace of A, on which A acts as a single eigenvalue. This consistency is exactly the projection postulate at work.

Commuting observables can be diagonalized *simultaneously* in one orthonormal basis, so a maximal set of commuting observables (a "complete set of commuting observables") labels states by joint eigenvalues—the quantum numbers of a system. Non-commuting observables share no eigenbasis, and the uncertainty relation of Lecture 9 measures how badly. The next lectures leave finite dimensions behind: dual spaces and tensors (Lectures 11–12), then Hilbert spaces and the spectral theorem for operators with continuous spectra (Lectures 13–14), where position and momentum finally find their home.

Example 10.50 (A complete set of commuting observables). On $\mathbb{C}^3$, $A = \operatorname{diag}(1,1,2)$ and $B = \operatorname{diag}(3,5,5)$ commute and share the standard orthonormal eigenbasis. Neither alone separates all three states—A cannot tell e_1 from e_2, B cannot tell e_2 from e_3—but the *joint* eigenvalues $(1,3), (1,5), (2,5)$ are distinct, so together A, B label the states completely, a finite model of a complete set of commuting observables. ◁

Example 10.51 (A shared eigenbasis). $A = \sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ commute, since $AB = BA = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, so they share the orthonormal eigenbasis $|\pm\rangle = \frac{1}{\sqrt{2}}(1,\pm 1)$: there $A = \operatorname{diag}(1,-1)$ and $B = \operatorname{diag}(3,1)$. Commuting observables diagonalize together even when neither is diagonal to begin with. ◁

Remark 10.52 (Density matrices). A statistical mixture of states is a *density matrix* $\rho = \sum_k p_k|\psi_k\rangle\langle\psi_k|$, a positive self-adjoint operator with $\operatorname{tr}\rho = 1$. By the spectral theorem ρ has an orthonormal eigenbasis with nonnegative eigenvalues summing to one—its own probability distribution over orthogonal pure states. The spectral theorem thus governs mixed states as well as pure ones.

Remark 10.53 (Beyond finite dimensions). The position operator has no eigenvectors in the usual sense: its spectrum is the *continuous* range of possible positions. The finite sum $A = \sum_k \lambda_k P_k$ then becomes an integral $\int \lambda\, dP(\lambda)$ against a projection-valued measure—the spectral theorem of Lecture 14, for which the finite-dimensional version proved here is the template.

Summary of main concepts

The spectral theorem says a normal operator on a complex space has an orthonormal basis of eigenvectors—equivalently a diagonal matrix in an orthonormal basis, $T = UDU^\dagger$ with U unitary. Self-adjoint operators are the special case with real eigenvalues (and a real orthogonal eigenbasis when the space is real), the principal-axes theorem. Grouping eigenspaces gives the spectral decomposition $T = \sum_k \lambda_k P_k$ with orthogonal projections satisfying $P_j P_k = \delta_{jk} P_k$ and $\sum_k P_k = I$, whence $f(T) = \sum_k f(\lambda_k) P_k$. In physics this is the statement that every observable has a complete orthonormal set of real-eigenvalue eigenstates; measurement projects onto them with Born probabilities, and commuting observables diagonalize together. Under a perturbation $H_0 + \varepsilon V$ the spectrum moves by at most $|\varepsilon|\|V\|$, and the shift of a simple level is computed order by order by Rayleigh–Schrödinger, each correction being a matrix element of V over a spectral gap; a degenerate level instead splits, by the eigenvalues of V compressed into its eigenspace.

- **Schur's theorem**—every complex operator is upper-triangular in some orthonormal basis.
- **Complex spectral theorem**—normal $\iff$ orthonormal eigenbasis $\iff$ unitarily diagonalizable ($T = UDU^\dagger$).
- **Real spectral theorem**—self-adjoint $\iff$ a real orthonormal eigenbasis (principal axes); real eigenvalues.
- **Spectral decomposition**—$T = \sum_k \lambda_k P_k$ with $P_j P_k = \delta_{jk} P_k$, $\sum_k P_k = I$; $f(T) = \sum_k f(\lambda_k) P_k$.
- **Operator norm and stability**—for normal T, $\|T\| = \max_k |\lambda_k|$ and $\|(T - zI)^{-1}\| = 1/\operatorname{dist}(z, \operatorname{spec} T)$; hence $\operatorname{spec}(H_0 + \varepsilon V)$ lies within $|\varepsilon|\|V\|$ of $\operatorname{spec}(H_0)$.
- **Rayleigh–Schrödinger**—$E_n^{(1)} = \langle n|V|n\rangle$, $E_n^{(2)} = \sum_{m\neq n} |\langle m|V|n\rangle|^2/(E_n - E_m)$; the expansion parameter is coupling over gap.
- **Degenerate perturbation**—a degenerate level splits by the eigenvalues of PVP on range P (self-study).
- **Physics**—observables have a complete orthonormal eigenbasis; Born-rule measurement; commuting observables diagonalize together.

Exercises for this lecture are in the companion sheet `exercises-10.pdf`.